



AI-Based Facial Recognition Authentication System

Multi-Layer AI Security · Biometric Verification · Real-Time Threat Detection

React · FastAPI · ArcFace · MobileNetV3 · EfficientNet-B4 · SQLite · JWT

Presentation Agenda

OVERVIEW

01 Problem Statement & Objectives

02 Proposed Solution & Architecture

03 Registration & Authentication Flows

04 AI Models – Detection, Recognition, Liveness

05 Deepfake Detection & Decision Engine

06 Security, Database & Results

07 Limitations, Future Scope & Conclusion

Problem Statement

PROBLEM

Traditional face login systems are vulnerable to:



Photo Spoofing

Printed photos or displays shown to the camera



Screen Replay

Mobile/monitor replay of a legitimate user's face video



Video Replay

Pre-recorded video injected as a live camera feed



Virtual Camera

Software-generated camera input bypassing physical sensors



AI-Generated Faces

Deepfake or face-swap synthetic faces that fool matching



Identity Lookalikes

Visually similar people whose embeddings may be close

Project Objectives

OBJECTIVES

PRIMARY OBJECTIVES

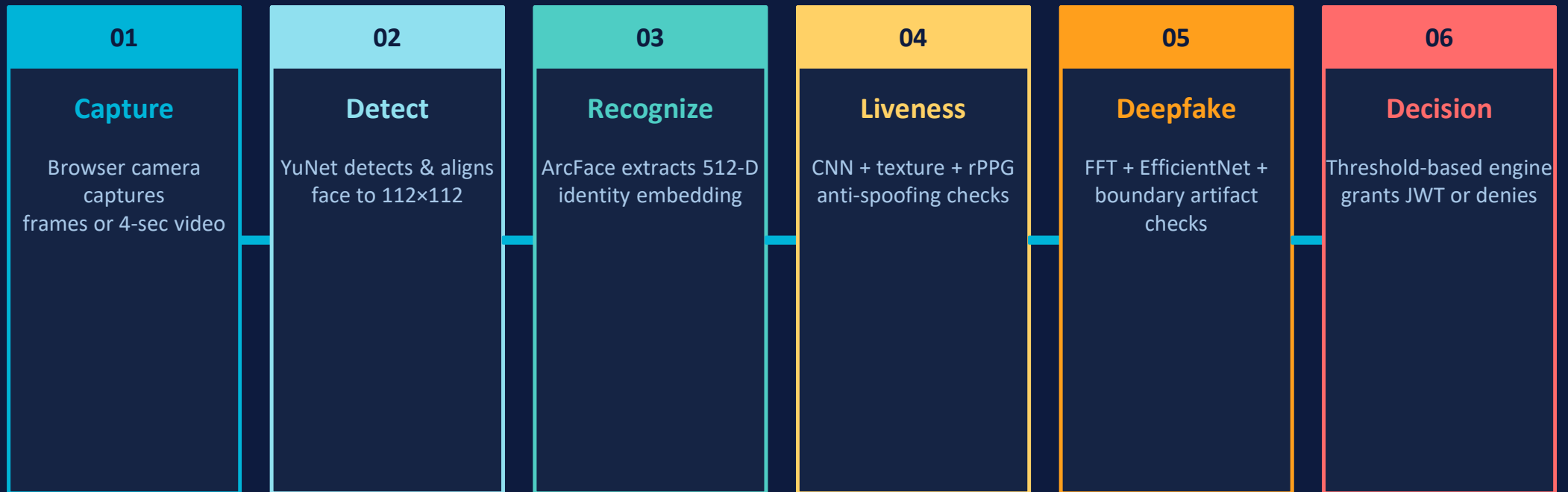
- Build a working face registration & authentication system
- Detect and align faces from browser-captured images/videos
- Extract robust 512-D embeddings using ArcFace
- Encrypt and securely store biometric templates (AES-256-GCM)
- Verify identity via cosine similarity against stored template
- Detect spoofing risk through liveness & anti-spoofing checks
- Detect deepfake risk using frequency, CNN, VLM texture & temporal cues
- Return clear grant/deny decisions with scores and threat flags
- Return clear grant/deny decisions with scores and threat flags

SECONDARY

- Clean web interface for registration & login
- Video-based auth for multi-frame analysis
- Optional instruction challenges
- Audit log of all attempts
- Training scripts for model fine-tuning
- Full architecture documentation

Proposed Solution

SOLUTION



GRANT — Issue RS256 JWT token



DENY — Log threat flags & reason

Proposed Solution

AI-Based Facial Authentication with Liveness & Deepfake Protection

Full-stack biometric login combining identity verification, anti-spoofing, and synthetic-media detection



Identity

ArcFace embeddings



Liveness

Multi-signal fusion



Deepfake

Spectral + CNN



VLM Reasoning

Visual explanation

System Architecture

ARCHITECTURE

FRONTEND

- React 18 + Vite
- Axios HTTP client
- CameraCapture component
- VideoRecorder component
- InstructionChallenge UI
- Result & Score display

BACKEND

- FastAPI + Uvicorn
- AuthPipeline orchestrator
- AntiInjectionGuard
- AES-256-GCM crypto
- RS256 JWT auth
- SlowAPI rate limiting

AI MODULES

- YuNet face detector
- ArcFace recognizer
- Liveness detector
- Deepfake detector
- MediaPipe challenge
- ONNX Runtime inference

DATA LAYER | SQLite · auth_logs · challenge_logs · Encrypted Embeddings | RS256 JWT Response

Technology Stack

TECH STACK

Frontend

React 18, Vite, Axios, React Router

Backend

FastAPI, Uvicorn, Python multipart

Computer Vision

OpenCV, NumPy, SciPy

AI Inference

ONNX Runtime, PyTorch, Torchvision

Face Detection

OpenCV YuNet ONNX model

Face Recognition

ArcFace w600k_r50.onnx

Liveness

MobileNetV3-Small, rPPG, optical flow

Deepfake

FFT, EfficientNet-B0, artifact checks

Challenge

MediaPipe FaceMesh & Hands

Database

SQLite, SQLAlchemy ORM

Security

AES-256-GCM, RS256 JWT, SlowAPI

Registration Flow

REGISTRATION



User enters username & email



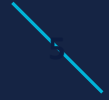
Frontend activates camera



Capture 5 JPEG face frames



POST /api/v1/register



YuNet detects & aligns each face



Quick liveness score per frame



ArcFace extracts embedding per frame



Average & L2-normalize template



Encrypt with AES-256-GCM



Store user in SQLite

AI Models Overview

AI MODELS

YuNet

FACE DETECTION

OpenCV-native ONNX
BBox + 5 landmarks
CPU-friendly

ArcFace

FACE RECOGNITION

w600k_r50.onnx
512-D embeddings
Cosine similarity

MobileNetV3-S

LIVENESS FEATURES

Torchvision pretrained
Texture & CNN features
Fine-tunable

rPPG

PULSE ESTIMATION

Green-channel signal
Forehead ROI analysis
Video-only

MediaPipe

CHALLENGE LANDMARKS

FaceMesh + Hands
Blink/gaze/gesture
Active liveness

EfficientNet-B0

DEEPPAKE FEATURES

ImageNet features
FFT + CNN hybrid
Boundary artifact checks

PROBLEM STATEMENT



Password Vulnerabilities

Passwords are stolen, guessed, phished, and reused — they offer no guarantee of real user presence.



Basic Face Systems Fail

Simple face-match systems are trivially defeated using printed photos, recorded videos, or 3D masks.



Rising Synthetic Threats

GAN-generated deepfakes and AI face-swaps can fool single-model recognition with near-perfect realism.

Modern authentication demands multi-layer intelligence — not a single check.

AI PIPELINE OVERVIEW



Layer 0

Anti-Injection Guard

ACTIVE



Layer 1

Face Detection & Align

ACTIVE



Layer 2

Face Recognition

ACTIVE



Layer 3

Liveness Detection

ACTIVE



Layer 4

Deepfake Detection

ACTIVE

⚡ DECISION ENGINE — GRANT or DENY based on cascading threshold checks

ANTI-INJECTION GUARD

Verifies that the image source is a genuine hardware camera — not a virtual device or injected stream.

OS Device Enumeration

Lists connected camera devices. Flags virtual drivers (OBS, ManyCam, XSplit) by name-matching known virtual device identifiers.

PRNU Noise Variance

Real camera sensors produce unique photo-response noise patterns. Synthetic or replayed streams show anomalously low variance.

Metadata Inspection

Checks EXIF data and stream metadata for inconsistencies that reveal screen-captured or digitally injected content.

Upload Bypass

When images are uploaded via HTTP (not live webcam), the guard is skipped — only live camera streams are validated.

FACE DETECTION & ALIGNMENT

YuNet Face Detector

ONNX

- Detects faces in input frames with high precision
- Extracts 5 facial landmarks (eyes, nose, mouth corners)
- Enables similarity-transform alignment before recognition
- Lightweight — runs efficiently on CPU without GPU
- Provides bounding box confidence for quality filtering
- Minimum confidence threshold: 0.90 required to proceed

Alignment Process

1

Frame Input

Raw webcam frame received from frontend multipart POST

2

Face Detection

YuNet locates all faces; highest-confidence bounding box selected

3

Landmark Extraction

5-point landmarks identified: left eye, right eye, nose tip, left mouth, right mouth

4

Similarity Transform

Affine transformation aligns face to 112×112 canonical pose for ArcFace

FACE RECOGNITION — ArcFace

512

Embedding
Dimensions

w600k

Training
Dataset

ResNet50

Backbone
Architecture

 ≥ 0.40

Similarity
Threshold

How Recognition Works

- Aligned 112×112 face image is passed through ArcFace ResNet-50
- A 512-dimensional L2-normalized embedding is produced
- During registration: 5 embeddings are averaged into one template
- Template is encrypted (AES-256-GCM) and stored in SQLite
- During auth: cosine similarity compared against stored template
- Similarity ≥ 0.40 required to proceed to final grant decision

FAISS Vector Index

- IndexFlatIP (inner product) used at dev/demo scale
- Enables sub-millisecond similarity lookup across users
- Supports future upgrade to IndexIVFFlat for >10K users
- Integrated with Python FAISS library for efficiency

DECISION ENGINE

IF	<code>injection.is_real_camera == False</code>	DENY	virtual_camera
IF	<code>detection.face_confidence < 0.90</code>	DENY	no_face
IF	<code>liveness.final_score < 0.80</code>	DENY	liveness_fail
IF	<code>deepfake.fused_score > 0.30</code>	DENY	synthetic_face
IF	<code>recognition.similarity < 0.40</code>	DENY	identity_mismatch
ELSE	ALL checks pass	GRANT	confidence = avg(liveness, similarity, 1-deepfake)

AUTHENTICATION FLOW

1

Enter Username

User provides username; system looks up the registered record

2

Capture Face Image

Camera captures a single login face image

3

Decrypt Template

Stored AES-256-GCM encrypted embedding is decrypted in memory

4

Anti-Injection Check

Layer 0 validates the camera source is genuine hardware

5

Face Detection

YuNet detects face; below 0.90 confidence → DENY

6

Recognition

ArcFace embedding computed; cosine similarity checked vs template

7

Liveness + Deepfake

Fused liveness ≥ 0.80 and deepfake ≤ 0.30 both required

8

Grant or Deny + Log

JWT RS256 returned on success; attempt logged in auth_logs table

THREAT MODEL COVERAGE

L0

Virtual Camera Injection

OS device enumeration + virtual driver name matching

L0 + L3

Replay Attack (pre-recorded video)

PRNU noise variance + rPPG pulse detection

L3

Printed Photo Attack

CNN anti-spoof + rPPG (no blood flow) + active challenge

L3

3D Silicone Mask

rPPG (no subcutaneous blood flow) + spectral analysis

L4

Deepfake Photo/Video (GAN/FLUX)

FFT spectral artifacts + EfficientNet-B4 classification

L4

AI Face-Swap Avatars (>30fps)

Spectral MLP detects GAN upsampling patterns

L2+L4

GAN Face (no real identity)

No FAISS match + spectral anomaly detection

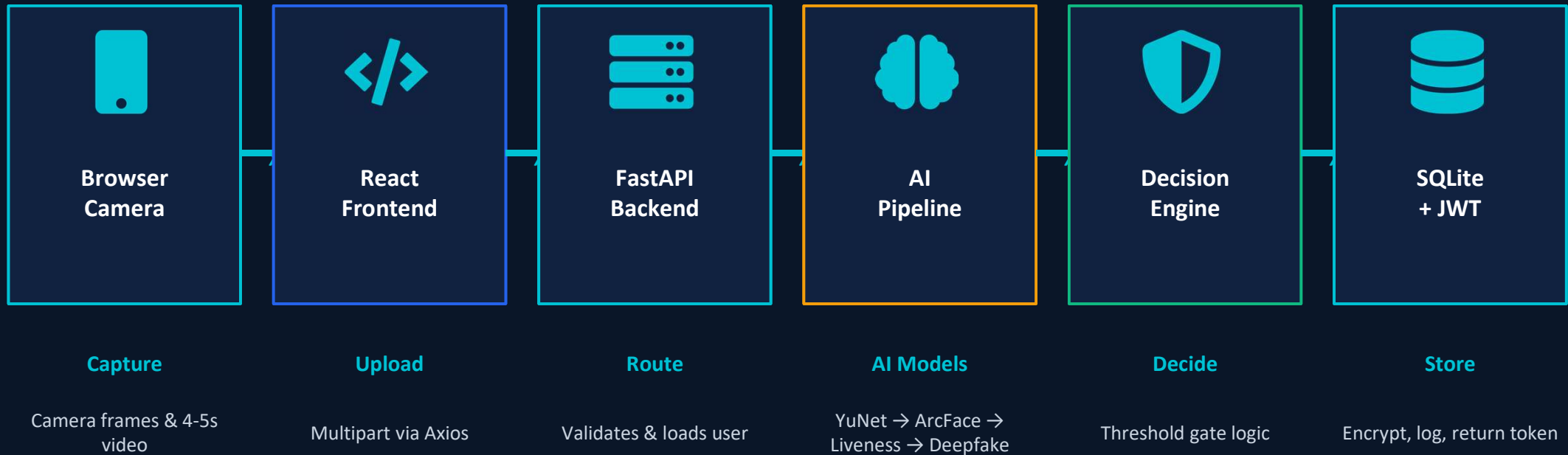
L4

Adversarial Perturbations

Spectral domain analysis is robust to pixel-space attacks

System Architecture

End-to-end request lifecycle



Two Authentication Tracks:

Traditional Pipeline (register + video login) | VLM Hybrid Pipeline (video register + VLM reasoning)

Technology Stack

Components powering the system

Frontend

- React 18 + Vite
- React Router
- Axios (multipart)
- navigator.mediaDevices
- MediaRecorder API

Backend

- FastAPI + Uvicorn
- Python 3.11/3.12
- SQLAlchemy + SQLite
- SlowAPI rate limiting
- RS256 JWT

AI / CV

- OpenCV YuNet ONNX
- ArcFace w600k_r50
- ONNX Runtime
- PyTorch + Torchvision
- MediaPipe FaceMesh

Security

- AES-256-GCM encrypt
- RS256 JWT issuance
- Audit logging
- Rate limiting
- CORS control

VLM Extension:

Qwen2.5-VL-3B-Instruct · moondream2 · Transformers · CUDA / CPU auto-select · psutil RAM checks

Registration Flow

Building a secure face template



Multiple frames produce a more stable, noise-resistant identity template than a single capture.

YuNet & ArcFace

Face detection, alignment, and identity embedding



YuNet — Face Detection

- OpenCV-native ONNX model
- Detects 5 facial landmarks
- Aligns face to 112×112 crop
- Rejects small / low-confidence faces
- Filters extreme yaw & pitch angles



ArcFace — Identity Embed

- w600k_r50.onnx model
- 512-dimensional embedding vector
- L2-normalized output
- Cosine similarity comparison
- Threshold: similarity ≥ 0.40

similarity = dot(auth_embedding, stored_template) → grant if ≥ 0.40

Liveness Detection

Multi-signal fusion to confirm a live capture



CNN Features

Unnatural texture statistics via MobileNetV3-Small



Texture

Flat print or screen surface patterns



Color Cues

Unnatural saturation and chrominance analysis



Moiré Pattern

Screen replay interference frequencies



Optical Flow

Static photos or unnatural inter-frame motion



Micro-movement

Absence of subtle live head motion



rPPG Signal

Pulse-like green-channel variation across frames



Challenge Score

Compliance with a fresh active instruction

Fusion Threshold: LIVENESS SCORE $\geq$ 0.70 → PASS

Deepfake Detection

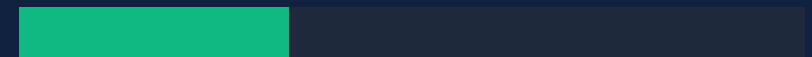
Estimating synthetic-media risk probability

Artifact Evidence Families

■ FFT Spectral	High-freq GAN anomalies
■ EfficientNet-B0	Deep CNN feature analysis
■ Boundary	Blending discontinuities
■ Eye Reflections	Inconsistent highlight patterns
■ Skin Uniformity	Over-smoothed texture regions
■ Color Correlation	Abnormal channel relations
■ Temporal Flicker	Frame-to-frame instability

Risk Probability

Safe



10% — clean capture

Caution



25% — near threshold

Flagged



35% — synthetic risk

Detection Threshold: DEEPPFAKE SCORE > 0.30 → DENY (synthetic_face)

Active Challenge System

Replay-resistant instruction verification

How It Works

■ GET /api/v1/challenge

Backend selects 2 random instructions after session starts

■ Video per instruction

User records one video response per action

■ MediaPipe FaceMesh

Verifies face actions — blink, turn, smile, raise brows

■ MediaPipe Hands

Verifies hand gestures — wave, open palm, touch nose

■ Instruction score fused

Challenge confidence included in liveness decision

Sample Instructions

- blink twice
- turn head left
- smile
- raise eyebrows
- show open palm
- wave hand

Why Challenges Help

- ✓ Printed photo cannot blink on command
- ✓ Old recorded video cannot match a fresh prompt
- ✓ Deepfake replay is harder with unpredictable actions

Model Roles

Each specialist focuses on a single security question

Anti-Injection

Is the camera source physical?

YuNet

Is there a usable face?

ArcFace

Is this the registered identity?

Liveness Fusion

Does the face appear live?

Deepfake Detector

Does the face look synthetic?

MediaPipe Challenge

Did the user respond to a fresh prompt?

VLM Reasoner

Is there visual consistency?

AES / JWT / Logs

Are data and decisions secured?

Strength comes from combining evidence, not from any single model.

Decision Engine

Sequential threshold gates — any failure = DENY

Gate 1: Camera Source

Pass: Anti-injection check

Deny: virtual_camera

Gate 2: Face Detection

Pass: Valid face required

Deny: no_face

Gate 3: Liveness

Pass: Score ≥ 0.70

Deny: liveness_fail

Gate 4: Deepfake

Pass: Probability ≤ 0.30

Deny: synthetic_face

Gate 5: Instructions

Pass: Required action passed

Deny: instruction_fail

Gate 6: Identity Match

Pass: Similarity ≥ 0.40

Deny: identity_mismatch

✓ All gates pass → GRANT + Issue
JWT

VLM Hybrid Authentication

Adding semantic visual reasoning to the pipeline

Why VLM?

- Numeric scores don't explain visual evidence
- VLM compares registration vs. auth frames
- Returns natural-language reasoning
- Adds identity, liveness, authenticity signals
- Runs only after traditional pipeline grants

VLM Response Shape

```
{ "same_person": true,  
  "same_person_confidence": 0.88,  
  "is_live": true, "liveness_confidence": 0.80,  
  "is_authentic": true, "overall_score": 0.85,  
  "reasoning": "...natural language..." }
```

VLM Pipeline

- Record 5-sec video
- Run traditional pipeline
- DENY? → skip VLM
- GRANT? → load ref frames
- VLM comparison
- Fuse scores (60/40 weight)
- Return decision + reasoning

VLM Registration Flow

Building a biometric profile with reference frames

1

Enter Details

Username & Email

2

Record 5-sec Video

Browser MediaRecorder

3

Decode Frames

OpenCV VideoCapture

4

Sample Frames

Distributed selection

5

Traditional Template

YuNet → ArcFace → AES encrypt

6

Select Best Frames

Quality-ranked reference set

7

Store Metadata

vlm_registrations table

8

JPEG on Disk

data/vlm_ref_frames/{user_id}/

Result: encrypted ArcFace embedding + VLM reference JPEG frames stored + vlm_registrations metadata

VLM Authentication Flow

POST /api/v1/vlm/authenticate

Backend Steps

- Load user & decrypt template
- Load VLM reference frames from disk
- Run traditional video pipeline
- Traditional DENY → skip VLM, return
- Traditional GRANT → extract frames
- VLM compare reg vs. auth frames
- Parse structured JSON judgment
- Fuse scores & check veto rule
- Return decision + JWT + reasoning

Response Fields

authenticated	true / false
confidence	fused 0.60 trad + 0.40 VLM
vlm_reasoning	natural language text
vlm_model_used	qwen2.5-vl-3b / moondream2
vlm_invoked	was VLM called?
vlm_override	did VLM veto?
scores.vlm_identity	same person confidence
scores.vlm_liveness	liveness confidence
scores.vlm_authenticity	authenticity confidence
jwt_token	RS256 on GRANT

VLM Decision Fusion

Score weighting and veto rules

$$\text{Final Confidence} = 0.60 \times \text{Traditional Confidence} + 0.40 \times \text{VLM Overall Score}$$



Traditional DENY

VLM is completely skipped. Denial reason and model scores are returned immediately. No VLM compute is wasted.



Traditional GRANT + VLM Agrees

Scores fused. Final confidence is weighted blend. JWT is issued. VLM reasoning returned in response.



Traditional GRANT + VLM Veto

If VLM deny confidence ≥ 0.85 and reports same_person=false / not live / not authentic → DENY override.

Fallback: If VLM cannot load or inference fails → neutral VLM result returned, traditional decision preserved

Security Features

Protecting biometric data and authentication decisions



AES-256-GCM Encryption

ArcFace embeddings are encrypted before SQLite storage. The symmetric key is injected via environment variable FACE_AUTH_AES_KEY.



RS256 JWT Tokens

On GRANT, the backend issues an RS256-signed JWT. Token is only returned after all security gates pass.



Rate Limiting

SlowAPI enforces 5 requests/minute on sensitive endpoints. Requests over limit receive HTTP 429.



Audit Logging

Every authentication attempt is written to auth_logs with decision, scores, threat flags, IP address, and denial reason.



CORS Control

Only local development origins are permitted. Production deployment requires proper CORS configuration.



Threat Flags

Detected anomalies are collected into a JSON threat-flag array and returned in every authentication response.

API Reference

Endpoints exposed by the FastAPI backend

Method	Endpoint	Purpose
GET	/health	Pipeline status check
POST	/api/v1/register	Register user from 5 face images
POST	/api/v1/authenticate	Legacy still-image auth
POST	/api/v1/authenticate/video	Current video login endpoint
GET	/api/v1/challenge	Issue a random instruction challenge
POST	/api/v1/authenticate/challenge	Auth with two instruction videos
GET	/api/v1/instructions	Full instruction catalog
GET	/api/v1/users/{id}/history	Last 10 auth attempts for user
POST	/api/v1/vlm/register	VLM registration with video
POST	/api/v1/vlm/authenticate	Traditional + VLM hybrid login
GET	/api/v1/vlm/status	VLM model & hardware readiness

Threat Scenarios

How layered gates defeat different attack vectors

Live User	Printed Photo	Screen Replay	Deepfake Face
Camera	Camera	Camera	Camera
Face	Face	Face	Face
Liveness	Liveness	Liveness	Liveness
Deepfake	Deepfake	Deepfake	Deepfake
Identity	Identity	Identity	Identity
GRANT	DENY: liveness_fail	DENY: liveness_fail	DENY: synthetic_face

Layered evidence means a different gate catches each distinct attack type.

Conclusion

A practical, explainable, layered approach to biometric authentication



Identity Verified

ArcFace cosine similarity against encrypted stored template



Liveness Confirmed

8-signal fusion detects photos, replays, and static input



Deepfake Screened

Spectral + CNN + boundary + temporal multi-evidence analysis



Data Secured

AES-256-GCM embeddings, RS256 JWT, audit logs



VLM Explainable

Natural-language reasoning on registration vs. auth frames



Extensible Platform

Modular design, evaluation scripts, and clear upgrade path

Stronger than face recognition alone · Auditable · Encrypted · Explainable